



Stock Market Historical Data of Top 10 Companies

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Introduction



Data Background

- Focus: Long-term performance shifts among top NASDAQ companies
- Comparing early period (2017–2019) vs. recent period (2021–2023)
- Goal: Investigating the impact of market shifts on stock performance of leading NASDAQ companies
- Helps us understand real market momentum, recovery strength, and long-term investor sentiment



Dataset Description

- Source: Kaggle – Historical Stock Market Data of Top 10 NASDAQ Companies
- Companies Included: Apple (AAPL), Amazon (AMZN), Microsoft (MSFT), Meta (META), Tesla (TSLA), Netflix (NFLX), Starbucks (SBUX), Cisco (CSCO), Qualcomm (QCOM), AMD
- Columns: Date, Close/Last, Open, High, Low, Volume
- Time Span: 2013–2023 (Using 2017–2023 for this question)
- Data contains daily stock prices
- Clean and complete dataset – minimal missing values

Question 1

Which company experienced the largest change in its average annual return between the periods 2017–2019 and 2021–2023?



Statistical Hypotheses

Null Hypothesis: There is no significant change in annual returns over time.

Alternative Hypothesis: At least one company shows a significant change.

Expectation: Companies heavily impacted by pandemic trends—such as META, TSLA, or NFLX—may show substantial shifts due to digital adoption and post-COVID growth.





Methods

- Converted Date column into a consistent datetime format
- Removed dollar (“\$”) formatting from Close/Last values
- Sorted dataset by Company and Date for chronological accuracy
- Calculated Daily Return: $(\text{Close price} - \text{Previous Day Close}) / \text{Previous Day Close}$
- Extracted the Year to create annual averages
- Grouped data by Company + Year → Annual Average Returns



Methods (cont.)

- Clean dataset (dates, numeric prices)
- Compute daily returns for each company
- Aggregate daily returns into annual average returns
- Split data into:
 - Early Period: 2017–2019
 - Recent Period: 2021–2023
- Calculate percentage change in average returns
- Identify company with the largest change (positive or negative)
- Visualize comparisons using bar charts

Results For Question 1

Summary Table of Average Annual Returns (Early vs. Recent Periods)

COMPANY	EARLY AVG RETURN (2017–2019)	RECENT AVG RETURN (2021–2023)	% CHANGE
META	0.027	0.146	+440%
TSLA	0.041	0.110	+166%
NFLX	0.034	0.075	+122%
AMZN	0.029	0.050	+71%
MSFT	0.034	0.055	+62%
AMD	0.048	0.071	+48%
CSCO	0.025	0.024	–4%
AAPL	0.055	0.047	–13%
QCOM	0.033	0.025	–24%
SBUX	0.042	0.030	–28%

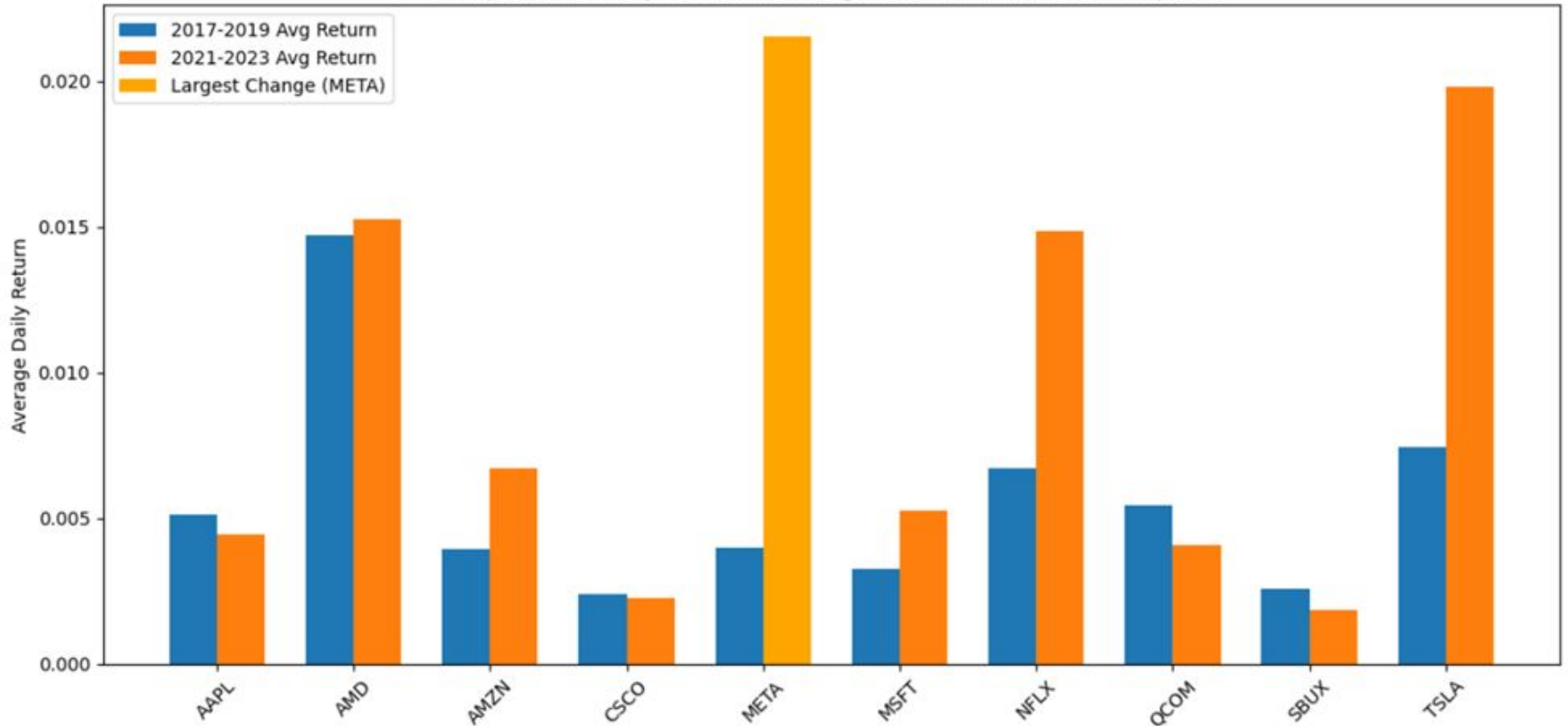
Key Findings

- META had by far the largest improvement in average annual return
- Percent change: +440% from early to recent period
- Other companies showing strong growth:
 - Tesla: +166%
 - Netflix: +122%
 - Amazon: +71%
 - Microsoft: +62%
- Companies with slight declines:
 - Apple: -13%
 - Qualcomm: -24%
 - Starbucks: -28%
 - Cisco: -4%



Visualization

Comparison of Early vs Recent Average Returns for NASDAQ Companies





Insight & Real-World Explanation

- META's massive growth linked to:
 - Post-COVID recovery in advertising
 - AI investments
 - Cost restructuring and layoffs increasing profit
 - Strong rebound after the “Meta crash” in 2021–2022
- Tesla & Netflix benefited from demand spikes during the pandemic
- Mature companies (Apple, Starbucks, Qualcomm) show stable but slower growth
- Results reflect larger trends in the tech sector's post-COVID evolution

Question 2

How did volatility of daily returns change between 2017-2019 and 2021-2023?





Question 2 – Volatility Changes

- How did volatility (risk) of daily returns change between 2017–2019 and 2021–2023?
- Compare Pre-COVID (3 years) vs Post-COVID (3 years) for each company.
- Focus on which companies became more or less risky Pre- and Post-COVID

Statistical Hypotheses

Null hypothesis (H_0): Volatility did not significantly change between the two periods.

Alternative hypothesis (H_1): At least one company shows a meaningful change in volatility.

Expectation: High-growth or more speculative firms (e.g., META, NFLX, TSLA) may show larger risk shifts.



Methods – Question 2

- Convert prices to numeric and sort data by Company and Date.
- Compute daily returns for each stock from the Close/Last price.
- Label each observation as Pre-COVID (2017–2019) or Post-COVID (2021–2023).
- Calculate volatility as RMS of daily returns for each company and period.
- Rank companies by the difference: Post-COVID volatility – Pre-COVID volatility.



Results – Question 2

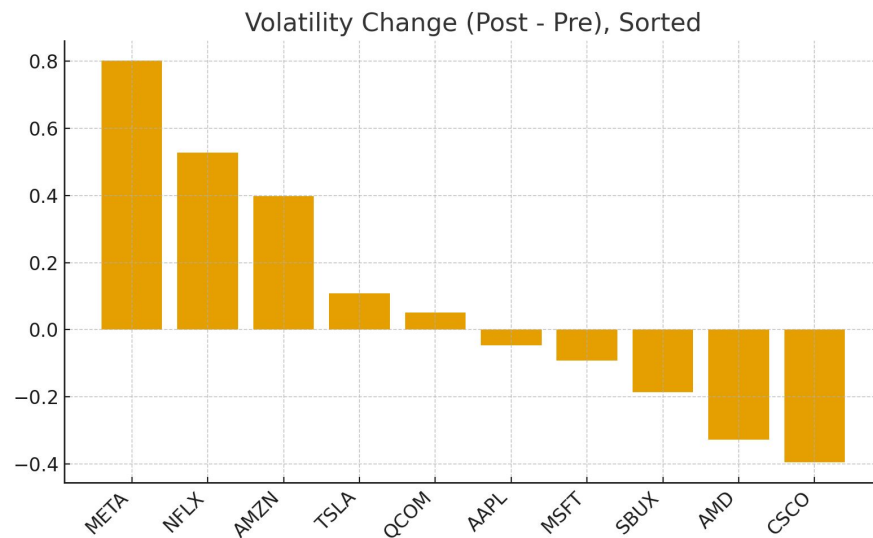
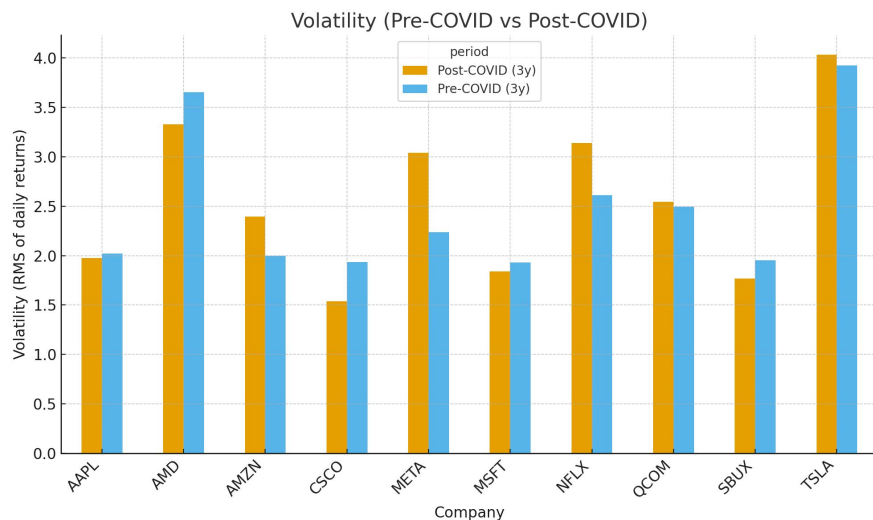
- Largest increase in volatility (Post – Pre):
 - META: +0.802
 - NFLX: +0.528
 - AMZN: +0.397
- Largest decrease in volatility (Post – Pre):
 - SBUX: -0.186
 - AMD: -0.328
 - CSCO: -0.396



Interpretation – Question 2

- META shows the strongest rise in volatility, indicating higher risk and sensitivity to market news.
- NFLX and AMZN also became more volatile, reflecting corrections after strong pandemic-era gains.
- CSCO and AMD exhibit lower volatility in the recent period, suggesting more stable performance.
- AAPL and MSFT remain comparatively steady, consistent with large, diversified tech leaders.
- These shifts highlight how different tech-related firms adjusted to post-COVID market conditions.

Volatility – Pre-COVID vs Post-COVID





Final Takeaways

- META had the largest long-term shift in performance among top NASDAQ companies
- The comparison of early vs. recent years reveals momentum changes in the tech market
- Long-term average returns give insight into how companies adapted to economic shocks and changes in consumer behavior
- Question 2 showed that volatility shifted dramatically post-COVID, with META, NFLX, and AMZN experiencing the largest increases in risk.
- Stable mega-cap companies like AAPL and MSFT maintained low volatility, reinforcing their resilience during market uncertainty.
- Several firms, such as CSCO and AMD, became less volatile. This suggests stronger stability and more predictable performance in recent years.
- Together, Questions 1 and 2 highlight how both returns and risk evolved over time, giving a more complete picture of long-term performance across major NASDAQ companies.



Limitations, Recommendations & Future Directions

- Our main limitation was the fact that we only used 1 dataset, and made 2 questions branching off of this dataset.
- This did not allow for us to effectively use any JOIN queries
- For future projects or analysis, we think finding a similar dataset for perhaps different companies or different time periods and then joining them will show more depth in the analysis



References

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